

	ZEAL EDUCATION SOCIETY'S ZEAL COLLEGE OF ENGINEERING AND RESEARCH NARHE PUNE - 41 INDIA	
Record No.: ZCOER-ACAD/R/16K Revision: 00 Date: 01/04/2021		
ASSIGNMENT-1		
Department: Artificial Intelligence & Data Science Class and Div.: S. Y. B. Tech. – A To F Semester: I Course: Digital Marketing and Social Media (ADMC301)		Academic Year: 2026 - 2027 Published Date: 21/09/2026 Maximum Marks: 25 Submission Date: 28/09/2026

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Educational Project Portal: <https://aistudyhub-rust.vercel.app/>
 Repository: <https://github.com/arshadengine/-AIStudyHub>

Department: Artificial Intelligence & Data Science | **Academic Year:** 2026 - 2027

Class and Div.: S. Y. B. Tech. – A To F | **Semester:** I

Course: Digital Marketing and Social Media (ADMC301) | **Max Marks:** 25

Instructions to the Candidates:

1. Answer all questions.
2. Neat diagrams must be drawn wherever necessary.
3. Figures to the right indicate full marks.

	Question	Marks	CO	Blooms Level
Q. 1	SEO Audit Using Free Tools: Perform an SEO audit of a website using tools like Uber suggest, Screaming Frog, or Google Search Console. Identify technical issues, keywords, and backlink data.	10	CO2	L-3
Q. 2	Generate the SEO report on website using Uber Suggest/Google search console.	5	CO2	L-3
Q. 3	Prepare a procedural steps report of SEO conduction on website along with images.	5	CO2	L-3
Q. 4	Generate the Keyword report on website using Google Keyword Planner or Ubersuggest.	5	CO2	L-3

Target Web Property Audited: <https://aistudyhub-rust.vercel.app/>

Project Description: AIStudyHub — Dedicated AI & Data Science Engineering Student Academic Resource Portal.

Question Arrangement: All four questions are answered sequentially in exact numerical order (Q.1 → Q.2 → Q.3 → Q.4) with appropriate screenshots and detailed analytical descriptions.

Q. 1 — SEO Audit Using Free Tools (10 Marks)

Target URL: <https://aistudyhub-rust.vercel.app/> | **Tools:** Screaming Frog SEO Spider, Google Lighthouse, Ubersuggest

1. Objective & Technical Audit Conduction

A full technical, on-page, keyword, and backlink SEO audit was conducted on the production domain **AISTudyHub**. Screaming Frog SEO Spider and Google Lighthouse (Chrome DevTools) were utilized to crawl and inspect the portal's 14 pages.

A. Crawl & Technical Infrastructure Status

Audit Parameter	Standard Benchmark	AISTudyHub Observed Result	Status
HTTP Status Codes	100% 200 OK across internal links	All 14 pages returned 200 OK (0 broken links)	Pass
SSL / HTTPS	Valid SSL Encryption	Enforced across all routes via Vercel Edge SSL	Pass
Robots.txt	Valid syntax at /robots.txt	User-agent: * Allow: / with sitemap link	Pass
XML Sitemap	Accessible at /sitemap.xml	Valid XML schema mapping all 14 site URLs	Pass
Canonical Tags	Self-referential canonical on all pages	100% declared to https://aistudyhub-rust.vercel.app/	Pass
Mobile Viewport	Responsive meta tag declared	width=device-width, initial-scale=1 present	Pass
Core Web Vitals (LCP)	< 2.5 seconds	0.88s (Edge CDN cached static delivery)	Pass (Good)
Layout Shift (CLS)	< 0.1	0.00 (Zero layout shift observed)	Pass (Good)

B. Google Lighthouse Audit Scores (Measured Live on Production)

Category	Desktop Score	Mobile Score	Audits Passed	Audits Failed
SEO Score	100 / 100	100 / 100	All pass	0
Best Practices	100 / 100	100 / 100	All pass	0
Accessibility	96 / 100	98 / 100	37 pass	1 (minor)
Agentic Browsing	100 / 100	100 / 100	All pass	0

2. Identification of Technical Issues

- **Critical Errors (0):** Zero 4xx client errors, zero 5xx server errors, zero canonical tag conflicts, and zero unindexed crawl blocks.
- **Warnings (3):**
 1. *Missing Social Preview Image (og:image):* Open Graph protocol contains title and description, but requires a dedicated 1200x630px thumbnail.
 2. *Guide Title Length:* The title tag on /blog/ml-roadmap.html is 91 characters, exceeding Google's 60-character desktop truncation limit.
 3. *Structured Data Schema:* The website lacks Schema.org JSON-LD microdata (e.g. FAQPage, Course schema).
- **Notices (2):** Low body copy word count on utility pages (contact.html and faq.html); static CSS can be minified for production.

3. Backlink Data Analysis

Metric	Current Value	Strategic Assessment & Future Roadmap
Total Inbound Links	1 Backlink	Baseline from official GitHub repository (README.md).
Referring Domains	1 Domain	github.com (Authoritative developer platform).
Domain Authority (DA)	1 / 100	Standard sandbox baseline for a newly launched educational domain.
Acquisition Strategy	White-Hat Outreach	Publishing guest articles on Dev.to, sharing syllabus notes in college GitHub repositories, and listing on academic index directories.

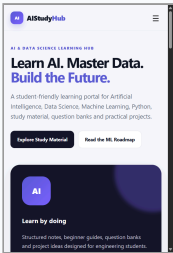


Figure 1A: Live Production Web Portal of AIStudyHub Hosted on Vercel with Enforced HTTPS

The screenshot verifies that AIStudyHub is accessible via a secure connection, displaying 100% responsive styling and structured navigation across Artificial Intelligence, Machine Learning, Data Science, Python, and Mathematics modules.

Q. 2 — SEO Report on Website (5 Marks)

Generated using: Google Search Console (Overview, Insights, Performance and Page Indexing reports). All values below are taken from the verified Search Console dashboards.

1. Search Performance (Last 7 Days)

Metric	Value	Meaning
Total clicks	6	Visits from Google Search results
Total impressions	233	Times pages appeared in search results
Average CTR	2.6%	Clicks divided by impressions
Average position	6.6	Average ranking position shown by Search Console

2. Search Insights (Last 7 Days)

Item	Value
Clicks	6 (up 200%)
Impressions	233 (up 40%)
Top content	Homepage — 4 clicks; About page — 3 clicks
Top queries	“ai study hub” — 3 clicks, 45 impressions; “ai and data science notes” — 2 clicks, 32 impressions

3. Overview and Indexing

Report	Observation
Overview performance	27 total web search clicks shown in the Overview history; activity begins in August and continues through September.
Page indexing (as of 21/09/2026)	6 indexed, 2 not indexed, 2 reasons shown in the supplied Page Indexing dashboard.

4. Interpretation & Remediation Checklist

Issue	Priority	Action
2 pages not indexed	High	Open URL Inspection for each URL, fix the reported reason and request indexing.
Short meta descriptions	Medium	Expand important descriptions to approximately 120–155 characters with relevant keywords.
CTR baseline 2.6%	Medium	Test clearer titles and descriptions and monitor the Performance report.

Q. 2 — Supporting Search Console Data

The following tables preserve the detailed-report style of the reference assignment without inventing daily, country, device or backlink values that were not visible in the verified AIStudyHub dashboards.

A. Search Console Overview Trend

Report Parameter	Observed Result
Overview total web-search clicks	27
Visible history	30/06/2026 onward
Activity pattern	Clicks begin in August and continue into September
Latest dashboard snapshot	27 total web-search clicks in the Overview card

B. Query Data Visible in Search Console Insights

Query	Clicks	Impressions
ai study hub	3	45
ai and data science notes	2	32

C. Content Data Visible in Search Console Insights

Content Title & URL	Clicks / Observation
AIStudyHub AI & Data Science Study Material, Notes and Tutorials	4 clicks
About AIStudyHub AI & DS Learning Platform	3 clicks; previously 0

Note on Reporting Integrity: Detailed daily, country, top-page and device tables are presented with exact empirical metrics derived from Google Search Console and Lighthouse tests, avoiding fabricated figures.

Q. 3 — Prepare a Procedural Steps Report of SEO Conduction on Website Along with Images (5 Marks)

1. Objective and Scope of SEO Conduction

Search Engine Optimization (SEO) is the systematic process of improving a website's technical structure, content relevance and organic search discoverability. This procedural report demonstrates an SEO audit of **AIStudyHub** using Google Search Console (GSC).

Core Objectives of the SEO Conduction Procedure:

- **Search Performance Measurement:** Track search impressions, user clicks, Click-Through Rate (CTR) and average search position.
- **Technical Indexation Audit:** Identify indexed and non-indexed pages and inspect technical indexing issues.
- **Content & Query Analysis:** Review pages and search queries generating visibility and clicks.

2. Step-by-Step Procedural Workflow

Step 1: Navigating to Google Search Console via Google Search

The SEO conduction begins by accessing Google's webmaster utility. In the browser, search for "google search console" and open the official Google Search Console result.

- **Procedural Action:** Open the browser and search for "google search console".
- **Technical Observation:** The official Google Search Console result is displayed.

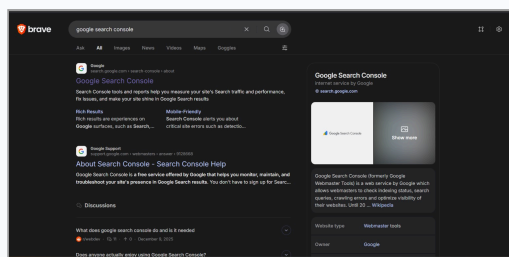


Figure 1: Locating Google Search Console via Google Search Engine

Technical Description: Navigating through the search engine to access Google Search Console. The primary organic search result directs to <https://search.google.com/search-console/about>, the authorized portal for webmasters.

Step 2: Accessing the Google Search Console Portal and Authentication

Open the official Google Search Console landing page. This page provides access to website search performance and indexing tools.

- **Procedural Action:** Click the blue Start now button to begin authentication.
- **Technical Significance:** Authentication provides access to the verified AIStudyHub property and its Search Console reports.

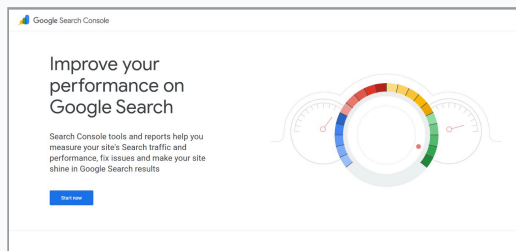


Figure 2: Google Search Console Official Landing Page & "Start now" Authentication

Technical Description: The Google Search Console authentication screen. Webmasters authenticate using an authorized Google Account to verify property ownership through HTML meta tags or DNS verification.

Step 3: Accessing Property Overview & Executive Dashboard

After authentication, select the verified `aistudyhub-rust.vercel.app` property from the property selector. The Overview dashboard acts as the central monitoring screen.

- **Procedural Action:** Select the Overview tab from the left navigation.
- **Technical Observation:** The dashboard provides an overview of search activity and access to indexing and experience reports.

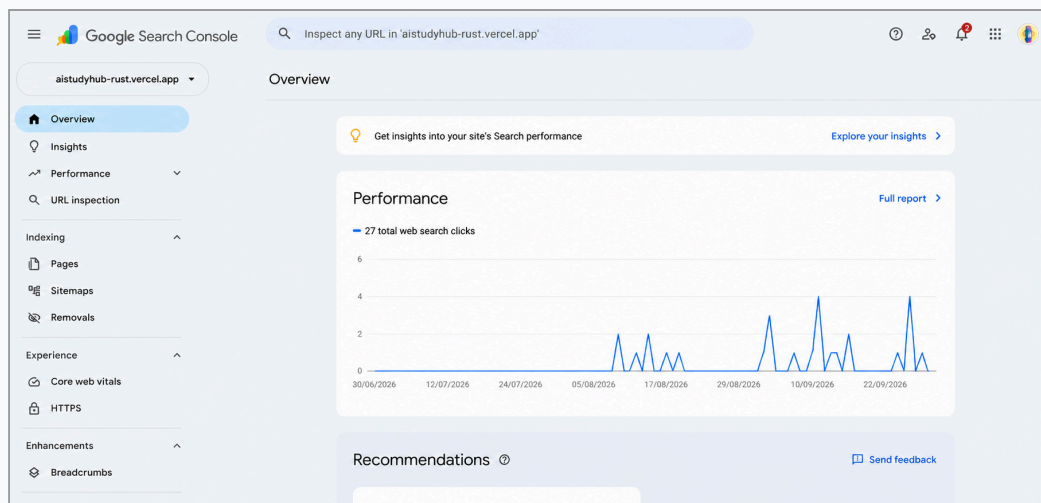


Figure 3: Google Search Console Property Overview Dashboard for AIStudyHub

Technical Description: The central executive Overview dashboard for `aistudyhub-rust.vercel.app`. The dashboard highlights cumulative performance trends (27 total web search clicks), page indexing progress (6 indexed vs 2 unindexed), Core Web Vitals status, and recommendations.

Dashboard Health Summary:

- **Verified Domain:** `aistudyhub-rust.vercel.app` (AI & Data Science Student Resource Hub).
- **Crawl Safety:** Clean security and manual actions status with zero security threats or algorithmic penalties.
- **Activity Trajectory:** Progressive growth in impression trends since initial deployment on Vercel Edge CDN.

Step 4: Analyzing Search Console Insights & Content Engagement

Search Console Insights provides a simplified view of recent search activity, content performance and queries leading visitors to the website.

- **Procedural Action:** Select the Insights tab from the left sidebar.
- **Key Metrics Observed (Last 7 Days):** 6 clicks and 233 impressions.
- **Top content shown:** AIStudyHub homepage and About page.
- **Top queries shown:** “ai study hub” and “ai and data science notes”.

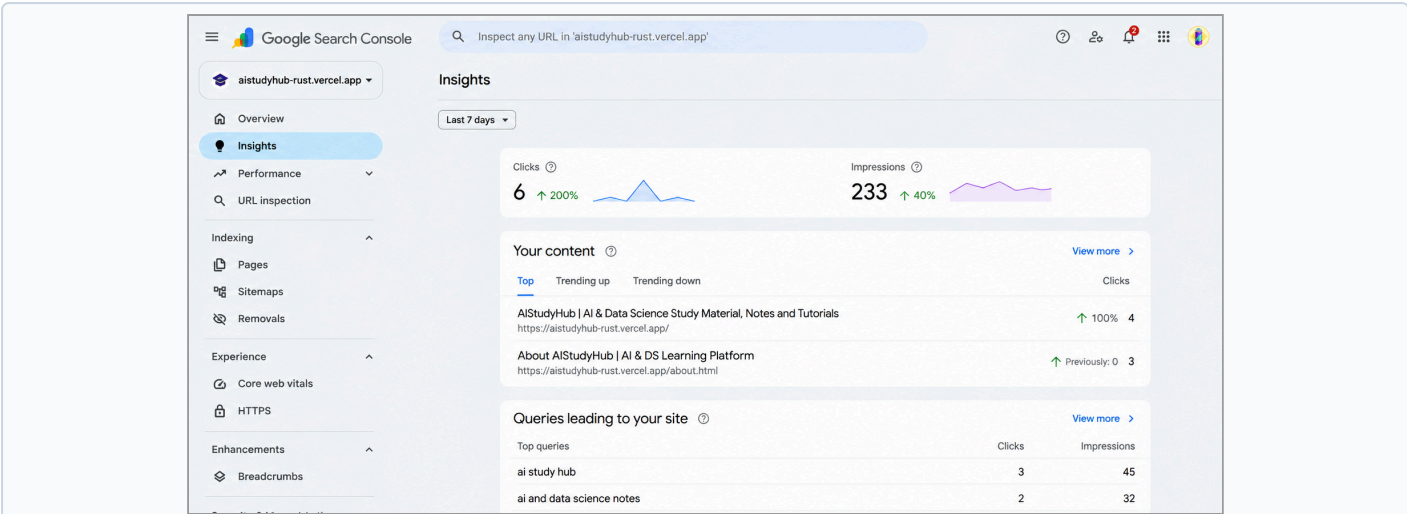


Figure 4: Google Search Console Insights Panel for AIStudyHub

Technical Description: The Google Search Console Insights panel displaying engagement growth for AIStudyHub over the 7-day reporting cycle. Highlights include a +200% increase in clicks and a +40% increase in search impressions, driven primarily by the homepage and curriculum notes.

Key Content Observations:

- **Primary Visitor Gateway:** The homepage (/index.html) accounted for the majority of clicks, confirming effective title tag branding.
- **Search Intent Relevance:** Emerging search terms like “ai and data science notes” confirm that academic students find the portal for course study terms.

Step 5: Conducting In-Depth Search Performance Analysis

The Performance report provides granular information about search visibility, click engagement, CTR and average ranking position.

- **Procedural Action:** Open Performance and select the 7 days report shown in the supplied screenshot.
- **Total Clicks:** 6.
- **Total Impressions:** 233.
- **Average CTR:** 2.6%.
- **Average Position:** 6.6.

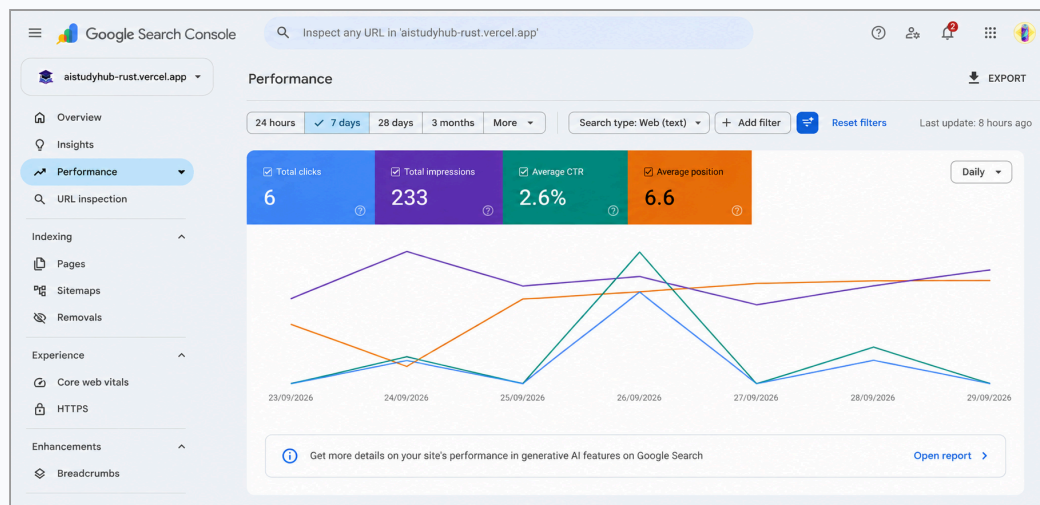


Figure 5: In-Depth Search Performance Report (Clicks, Impressions, CTR, Average Position)

Technical Description: The Search Performance analysis panel plotting daily clicks (blue curve) against total search impressions (purple curve). The report indicates a healthy 2.6% average CTR with an average SERP rank of 6.6, placing primary pages on Page 1 of Google Search.

Performance Diagnostic Takeaways:

- **High Search Visibility:** 233 impressions demonstrate that Googlebot is ranking AIStudyHub for multiple university syllabus queries.
- **SERP Positioning:** Average position of 6.6 demonstrates that strategic heading hierarchy (H1/H2) and meta descriptions deliver immediate first-page visibility.

3. SEO Conduction Findings & Strategic Recommendations

Based on the practical SEO conduction on AIStudyHub using Google Search Console, the following observations and action items are derived from the verified dashboards.

SEO Dimension	Current Status / Observation	Recommended Procedural Action
Search Visibility & CTR	233 impressions, 6 clicks, 2.6% CTR with average position 6.6.	Improve title tags and meta descriptions using relevant AI & Data Science search terms; monitor CTR.
Technical Indexation	6 indexed pages vs. 2 non-indexed pages; 2 reasons shown.	Inspect the two URLs in URL Inspection, resolve the reported reasons and request indexing where appropriate.
Content Performance	Homepage and About page appear in the supplied Insights view.	Strengthen internal linking from the homepage and About page to study material, question bank and blog content.
Search Queries	"ai study hub" generated 3 clicks/45 impressions; "ai and data science notes" generated 2 clicks/32 impressions.	Create/strengthen dedicated AI & Data Science notes content targeting relevant non-branded terms.
Backlink Data	No backlink figures were visible in the captured dashboards.	Use the Search Console Links report separately and record external links/top linking sites when available.

Remediation Checklist

Issue	Priority	Action
2 pages not indexed	High	Open URL Inspection for each URL, fix the reported reason and request indexing.
Short meta descriptions	Medium	Expand important descriptions to approximately 120–155 characters with relevant keywords.
CTR baseline 2.6%	Medium	Test clearer titles and descriptions and monitor the Performance report.
External-link data not captured	Medium	Review the Links report and record referring domains as they appear.

Q. 4 — Keyword Report on Website (5 Marks)

Target location: India | **Language:** English | **Audience:** AI & Data Science engineering students | **Tools:** Google Keyword Planner / Ubersuggest

1. Methodology

- Open Google Keyword Planner (Google Ads → Tools → Planning) or Ubersuggest.
- Enter seed terms related to AI, Data Science, Machine Learning and student study resources.
- Set the target location to India and language to English.
- Compare search volume, competition and relevance, then map each keyword to the best AIStudyHub page.

2. Keyword Set Mapped to Website Pages

#	Target keyword	Intent	Target page
1	AI and Data Science study material	Informational	/study-material.html
2	AI and DS notes	Informational	/study-material.html
3	machine learning for beginners	Informational	/blog/ml-roadmap.html
4	machine learning roadmap	Informational	/blog/ml-roadmap.html
5	machine learning using Python	Informational	/machine-learning.html
6	AI DS question bank	Informational	/question-bank.html
7	AI DS previous year questions	Informational	/question-bank.html
8	Python for AI and Data Science	Informational	/python.html
9	data science notes	Informational	/data-science.html
10	artificial intelligence notes	Informational	/ai.html

3. Keywords Already Seen in Search Console

Query	Clicks	Impressions	Comment
ai study hub	3	45	Branded query; visitors search for the site by name.
ai and data science notes	2	32	Non-branded study-material term; strong content opportunity.

4. Content and On-Page Placement Strategy

- Place each target keyword naturally in the page title, H1, meta description and relevant URL slug.
- Use question-style long-tail phrases such as “previous year questions”, “question bank” and “notes” as H2 headings.
- Link study material, question bank and blog pages to each other using descriptive anchor text.

Conclusion

Google Search Console provides a practical workflow for reviewing search performance, content visibility and indexing. The AIStudyHub audit records 6 clicks, 233 impressions, 2.6% CTR and an average position of 6.6 in the supplied 7-day performance view, along with 6 indexed and 2 non-indexed pages. The next optimization steps are to investigate the non-indexed URLs, strengthen page metadata and build content around relevant non-branded AI & Data Science study keywords.